

Concept 3 | Applications of Differentiation: Polynomial Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Stationary

$$f'(x) = 0$$

Second derivative

$$f''(x) > 0 \Rightarrow \text{minimum}$$

Rate

$$\frac{dy}{dx}$$

Eng Eslam Ahmed | 01120009622

Classified Practice

Applications of Differentiation: Polynomial Functions

Q001 **Written**

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

Q002 Written**Find the extreme values and sketch the three source Try cubics.**

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Q003 **Written**

Find the extreme values and draw the graphs for all nine source cubic functions in Exercise.

Q004 Written

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

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Q005 Written

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

Q006 Written**Find the maximum and minimum values of the two source Try cubics on their stated closed intervals.**

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Q007 **Written**

Find the maximum and minimum values of all six source cubics on their given closed intervals.

Q008 Written

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

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Q009 Written

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a, b , and the local minimum.

Q010 Written

Solve the two source Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

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Q011 Written

Solve all six source Exercise problems asking for cubic coefficients from extrema.

Q012 Written

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a, b, c .

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Q013 Written

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k .

Q014 Written

Determine the parameter ranges in the source Try problems for a cubic to have no extreme values or to have both extremes.

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Q015 **Written**

Determine the parameter ranges for all six source Exercise cubics according to whether they have extreme values.

Q016 Written

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, find the exact k -range that guarantees both a local maximum and minimum.

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Q017 Written

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k .

Q018 Written

Find the parameter ranges in the two source Try problems for the cubic to be monotonically increasing.

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Q019 Written

Find the parameter ranges in all six source Exercise problems for monotonic increase.

Q020 Written

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k + 6)x + 10$, find the k -range for monotonic increase on $x \in [1, \infty)$.

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Q021 Written

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.

Q022 Written**Find the number of distinct real roots for the three source Try equations.**

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Q023 Written

Find the number of distinct real roots for all nine source cubic equations.

Q024 Written**Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.**

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Q025 Written

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a .

Q026 Written**Solve the two source Try parameter problems for three or two distinct real roots.**

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Q027 **Written**

Solve all four source Exercise parameter problems for the required number of distinct roots.

Q028 Written

For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.

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Q029 Written

Prove the source inequalities using an auxiliary cubic and a derivative sign table.

Q030 Written**Prove all three source cubic inequalities on their stated domains.**

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Q031 Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ for every $x \in [-1, 3]$.

Q032 Written**Find the extreme values and sketch the two source quartic functions.**

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Q033 Written

Find the extreme values and graphs for all six source quartic functions.

Q034 Written

For $f(x) = x^4 - 4x^3 + k$, find the exact k-range for exactly two distinct real roots.

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Q035 MCQ

At a local maximum, the sign of f' changes:**Choose the correct option.**A $- \rightarrow +$ B $+ \rightarrow -$ C $+ \rightarrow +$ D $- \rightarrow -$

Q036 MCQ**For $P'(x) = 3[(x - a)^2 - 1]$, the local maximum occurs at:****Choose the correct option.****A** $a - 1$ **B** a **C** $a + 1$ **D** $1 - a$

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Q037 MCQ

Absolute extrema on a closed interval require comparing:**Choose the correct option.**

- A stationary points only
- B endpoints only
- C stationary points and endpoints
- D the y-intercept

Q038 MCQ

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:**Choose the correct option.**A $4\sqrt{2}$

B 9

C 6

D 0

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Q039 MCQ

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

Choose the correct option.

A 2

B 6

C 12

D -12

Q040 MCQ**The derivative roots in the challenge are:****Choose the correct option.**A $-4, 2$ B $-2, 4$ C $2, 4$ D $-3, 3$

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Q041 MCQ

No two distinct extrema for a cubic means the derivative discriminant is:**Choose the correct option.**A $D > 0$ B $D = 1$ C $D \leq 0$ D $D < 0$ only

Q042 MCQ

The challenge range for both extrema is:**Choose the correct option.**

A $0 < k < 4$

B $k < 0$ or $k > 4$

C $-4 < k < 0$

D $k \leq 4$

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Q043 MCQ

For a positive-leading cubic to be monotone increasing, its derivative must be:

Choose the correct option.

- A negative everywhere
- B non-negative everywhere
- C zero everywhere
- D linear

Q044 MCQ

The challenge domain in 7-13 is:**Choose the correct option.**A $(-\infty, \infty)$ B $[1, \infty)$ C $[-1, 1]$ D $(0, 1)$

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Q045 MCQ

Three distinct roots occur when the x-axis lies:**Choose the correct option.**

- A above both extrema
- B between the local maximum and minimum
- C at an endpoint only
- D at the local maximum only

Q046 MCQ**The 7-14 challenge has how many distinct real roots?****Choose the correct option.****A** 0**B** 1**C** 2**D** 3

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Q047 MCQ

Three intersections with $y = a$ require the level to lie:**Choose the correct option.**

- A strictly between the extrema
- B above the maximum
- C below the minimum
- D at an extremum

Q048 MCQ

The 7-15 warm-up range is:**Choose the correct option.**

A $-27 < a < 5$

B $-5 < a < 27$

C $a \leq -27$

D $a \geq 5$

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Q049 MCQ

To prove $LHS \geq RHS$, define:**Choose the correct option.**

A $f = LHS + RHS$

B $f = LHS - RHS$

C $f = RHS - 1$

D $f = f'$

Q050 MCQ

The largest k in the challenge is:**Choose the correct option.**

A $2 - 4\sqrt{2}$

B $4\sqrt{2} - 2$

C $2 + 4\sqrt{2}$

D $-2 - 4\sqrt{2}$

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Q051 MCQ

A repeated root of a derivative may be:**Choose the correct option.**

- A always a local maximum
- B always a local minimum
- C stationary but not an extreme
- D impossible

Q052 MCQ

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require:**Choose the correct option.**A $k = 27$ B $k < 27$ C $k > 27$ D $k \leq 0$ only

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Q053 MCQ

A local minimum is identified by the change:**Choose the correct option.**A $+$ $\rightarrow$ $-$ B $-$ $\rightarrow$ $+$ C $+$ $\rightarrow$ $+$ D $-$ $\rightarrow$ $-$

Q054 MCQ**For $y = x^3 - 6x^2 + 16$, the critical x-values are:****Choose the correct option.****A** 0, 4**B** 1, 3**C** -2, 2**D** 2, 4

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Q055 MCQ

The derivative sign table is used to determine:**Choose the correct option.**

- A only the y-intercept
- B increase/decrease and extrema
- C the constant term
- D the domain

Q056 MCQ**On a closed interval, an endpoint can be:****Choose the correct option.**

- A ignored
- B an absolute extreme
- C a derivative root only
- D never a minimum

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Q057 MCQ

For $y = -x^3 + 12x + 5$, the stationary x-values are:

Choose the correct option.

A $-2, 2$

B $-1, 1$

C $0, 4$

D $2, 4$

Q058 MCQ

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at:

Choose the correct option.

A $x = -3$

B $x = -\sqrt{2}$

C $x = \sqrt{2}$

D $x = 3$

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Q059 MCQ

To determine a cubic from an extreme value, use both $f(a) = M$ and:

Choose the correct option.

A $f(a) = 0$

B $f'(a) = 0$

C $f''(a) = 0$

D $a = 0$

Q060 MCQ**In the warm-up $f(x) = -x^3 + 12x - 7$, the local minimum occurs at:****Choose the correct option.**

A $x = -2$

B $x = 0$

C $x = 2$

D $x = 7$

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Q061 MCQ

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires:

Choose the correct option.

A $0 \leq k \leq 1/3$

B $k > 1/3$

C $k < 0$

D $k \geq 1/3$

Q062 MCQ**Two distinct stationary points occur when the derivative discriminant is:****Choose the correct option.**

A $D < 0$

B $D = 0$

C $D > 0$

D $D = 1$

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Q063 MCQ

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

Choose the correct option.

A non-negative for all x

B negative for all x

C linear

D undefined

Q064 MCQ

In the 7-13 challenge, monotonicity is required on:**Choose the correct option.**A $[1, \infty)$ B $[-1, 1]$ C $(-\infty, 0)$ D $[0, 1]$

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Q065 MCQ

A cubic with both stationary values above the x-axis has:**Choose the correct option.**

- A three roots
- B one distinct real root
- C no real roots
- D two repeated roots

Q066 MCQ**To count cubic roots, compare the extrema with:****Choose the correct option.**

- A the y-axis
- B the x-axis
- C the derivative axis
- D the domain only

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Q067 MCQ

At an extremal parameter level, a cubic and a horizontal line have:**Choose the correct option.**

- A one distinct intersection
- B two distinct intersections
- C three distinct intersections
- D no intersection

Q068 MCQ**For $x^4 - 4x^3 + k$, the global minimum occurs at:****Choose the correct option.****A** $x = 0$ **B** $x = 1$ **C** $x = 3$ **D** $x = 4$

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Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 1 Concept Quiz · Written

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

Quiz WRITTEN 2 Concept Quiz · Written

Find a, b if $f(x) = -x^3 + ax + b$ has a local maximum 4 at $x=1$.

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Quiz WRITTEN 3 Concept Quiz · Written**Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.**

Quiz WRITTEN 4 Concept Quiz · Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ on $[-1, 3]$.

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Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)**At a local minimum, f' changes:****Choose the correct option.**A $+$ $\rightarrow$ $-$ B $-$ $\rightarrow$ $+$ C $+$ $\rightarrow$ $+$ D $-$ $\rightarrow$ $-$

Quiz MCQ 6 Concept Quiz · MCQ

A cubic with a positive-leading derivative and $D < 0$ is:

Choose the correct option.

- A always increasing
- B always decreasing
- C constant
- D undefined

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Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

The 7-15 challenge requires the level to satisfy:

Choose the correct option.

A $t < 0$

B $0 < t < 4$

C $t > 4$

D $t = 4$

Quiz MCQ 8 Concept Quiz · MCQ

The global minimum of $x^4 - 4x^3 + k$ is:

Choose the correct option.

A $k - 27$

B $k + 27$

C $k - 3$

D $27 - k$

Answer key

Use the question number shown on each page.

- Q001** Use the derivative sign table: for $x^3 - 6x^2 + 16$, local maximum 16 at $x = 0$ and local minimum -16 at $x = 4$; classify the other two from their stationary points.
- Q002** Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.
- Q003** Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.
- Q004** Local maximum at $x = a - 1$; local minimum at $x = a + 1$.
- Q005** Maximum 21 at $x = 2$; minimum -11 at $x = -2$.
- Q006** Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.
- Q007** For each function, compare the endpoint values with the values at all critical x-values lying in the interval.
- Q008** Absolute minimum -9 at $x = -3$; absolute maximum 9 at $x = 3$.
- Q009** $a = 12, b = -7$; local minimum -23 at $x = -2$.
- Q010** Use $f'(a) = 0, f(a) = M$, and the second stationary condition to form three simultaneous equations in the coefficients.
- Q011** Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.
- Q012** $a = -3, b = -24, c = 150$.
- Q013** $0 \leq k \leq \frac{1}{3}$.
- Q014** Use $D \leq 0$ for no extremes and $D > 0$ for two distinct stationary points.
- Q015** Apply the derivative-discriminant criterion separately to each printed cubic.
- Q016** $k < 0$ or $k > 4$.
- Q017** $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$.
- Q018** Set the derivative discriminant $D \leq 0$, with the derivative leading coefficient positive.
- Q019** Differentiate each cubic and impose that its quadratic derivative never becomes negative.
- Q020** $k \leq 3$.
- Q021** Three distinct real roots.
- Q022** Use the cubic graph: count x-axis intersections from the stationary points and end behavior.
- Q023** Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.
- Q024** One distinct real root.
- Q025** $-27 < a < 5$.
- Q026** Rewrite each equation as $g(x) = a$, then compare the parameter line with the local maximum and minimum of g .
- Q027** Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.
- Q028** $\frac{7 - \sqrt{65}}{4} < a < \frac{7 - \sqrt{33}}{4}$ or $\frac{7 + \sqrt{33}}{4} < a < \frac{7 + \sqrt{65}}{4}$.
- Q029** Set $f(x) = \text{LHS} - \text{RHS}$, prove $f(x) \geq 0$ on the stated domain, and identify its minimum.
- Q030** Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.
- Q031** $k \leq 2 - 4\sqrt{2}$; the largest value is $2 - 4\sqrt{2}$.
- Q032** For $3x^4 + 4x^3 - 12x^2$: local maximum 0 at $x = 0$, local minima -32 at $x = -2$ and -5 at $x = 1$. For $-x^4 - 4x^3 - 8$: maximum 19 at $x = -3$; $x = 0$ is stationary but not an extreme.
- Q033** Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.
- Q034** $k < 27$.
- Q035** B
- Q036** A
- Q037** C
- Q038** B
- Q039** C
- Q040** B
- Q041** C
- Q042** B
- Q043** B
- Q044** B
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** B
- Q050** A
- Q051** C
- Q052** B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 A

Q058 A

Q059 B

Q060 A

Q061 A

Q062 C

Q063 A

Q064 A

Q065 B

Q066 B

Q067 B

Q068 C

Quiz WRITTEN 1 Maximum 1 at $x = 0$; minimum -3 at
 $x = 2$

Quiz WRITTEN 2 $a = 3, b = 2$

Quiz WRITTEN 3 $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$

Quiz WRITTEN 4 $2 - 4\sqrt{2}$

Quiz MCQ 5 B

Quiz MCQ 6 A

Quiz MCQ 7 B

Quiz MCQ 8 A